

EXPLAINABLE CNN FOR LUNG CANCER DETECTION FROM STANDARDIZED CHEST X-RAYS USING MULTI-LEVEL ATTENTION AND CLINICAL FEATURE FUSION

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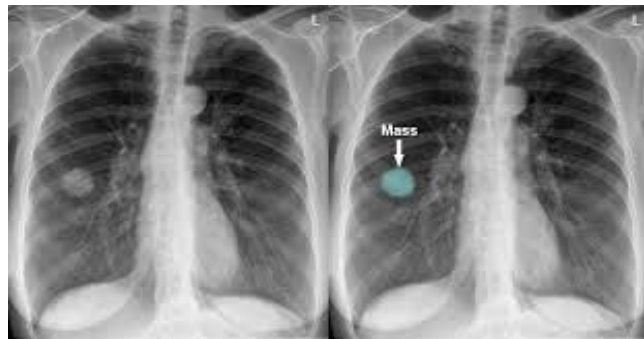
Date: 5/20/26

Computer Systems Lab, Yilmaz-p7

BACKGROUND

PROBLEM

- Lung cancer is often diagnosed too late because subtle early tumors are hard to detect on chest X-rays; ~20–30% of cases are missed, especially in early stages[11].
- Radiologists struggle to distinguish early cancer from normal anatomy, leading to false negatives; chest X-rays have only ~70–80% sensitivity, contributing to frequent misdiagnoses[11].



PROBLEM

- Existing CNN-based detection models often function as “black boxes,” giving predictions without explaining what visual patterns influenced their decision[10].
- This lack of transparency and reliability in current AI systems limits clinical trust and prevents their effective use in real-world lung cancer diagnosis[11].

BACKGROUND

- Lung cancer is the leading cause of cancer-related deaths worldwide[11].
- Early detection improves survival, but early-stage tumors are difficult to identify on chest X-rays due to low contrast and overlapping tissues; sensitivity is ~60-70% for early-stage cases[11].
- Chest X-rays remain the most widely used and affordable imaging method for lung screening, especially in low-resource settings[12].
- Standardized chest X-ray datasets provide large-scale, clinically representative images used to train and evaluate detection models[11].



BACKGROUND

- **Convolutional Neural Networks (CNNs)** are deep learning models that detect patterns in medical images and can identify subtle features missed by radiologists[13].
- CNNs learn hierarchical features (edges → structures), enabling detection of early-stage abnormalities in chest X-rays[13].
- **Clinical Feature Fusion** combines imaging outputs with patient data (e.g., age, smoking history), which is clinically important because lung cancer risk depends on both imaging findings and patient context, improving diagnostic accuracy[13].

OTHER SOLUTIONS

OTHER SOLUTIONS

CNN-based CAD system [1] for lung cancer detection using JSRT dataset of chest xray images, 154 cancerous/93 normal.

Automated lung cancer diagnosis from chest X-ray images using convolutional neural networks

[Aya Aboelghiet](#)¹, [Samaa M Shohieb](#)^{1,✉}, [Amira Rezk](#)¹, [Ahmed Abou Elfetouh](#)^{1,2}, [Ahmed Sharaf](#)^{3,✉}, [Islam Abdelmaksoud](#)¹

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PMCID: PMC12453845 PMID: [40989445](#)

Results:

Accuracy $83.2 \pm 2.91\%$, Sensitivity 83.7%, Specificity 82.8%, F1-score 79.4%.

Limitations:

Dataset is very small (247 images total), limiting generalizability to real clinical settings.

Not representative of diverse patient populations or real-world variability.

Risk of overfitting due to limited data.

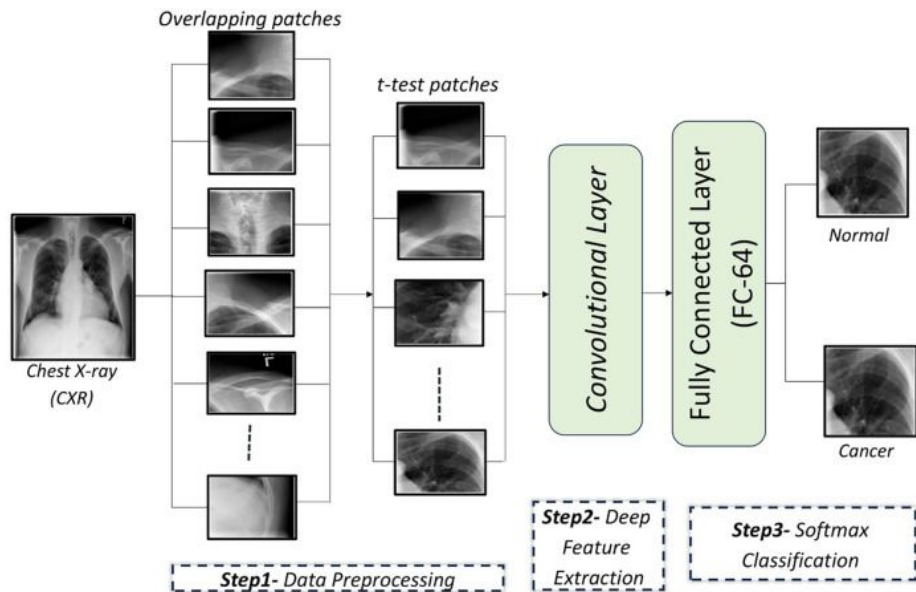
OTHER SOLUTIONS

CNN-based CAD system [1]

Methods:

Images were divided into overlapping lung patches, filtered using a t-test to retain informative regions.

A custom 7-layer CNN (3 conv, 2 pooling, 2 fully connected) was trained with ReLU activation and softmax output, optimized via stochastic gradient descent and 10-fold cross-validation [1].



CNN architecture used by [1].

OTHER SOLUTIONS

Uses Sparse Convolutional Neural Network (SCNN) for lung cancer detection using histological image classification [2], instead of chest x-rays.

Article | [Open access](#) | Published: 01 September 2025

Improving lung cancer detection with enhanced convolutional sequential networks

[Usman Haziq](#), [Jamal Uddin](#), [Shahid Rahman](#), [Muhammad Yaseen](#), [Inayat Khan](#), [Jawad Khan](#) ✉ & [Younhyun Jung](#) ✉

[Scientific Reports](#) **15**, Article number: 32099 (2025) | [Cite this article](#)

2059 Accesses | **1** Altmetric | [Metrics](#)

Accuracy 95.34%, Recall 95.33%, F1-score 95.66%.

Outperforms CNN, because they are not using chest rays as a sample data and are using histopathological dataset of digitized tissue images.

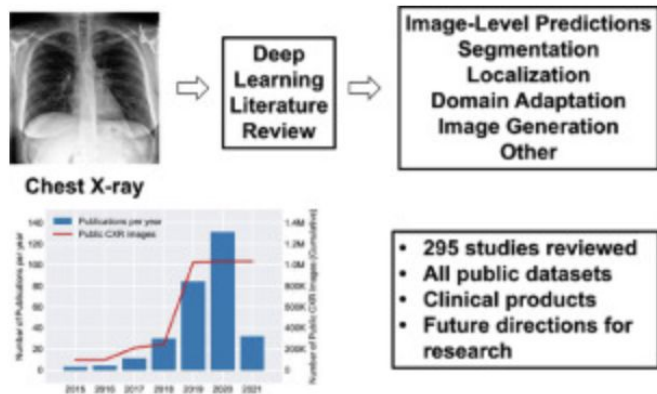
OTHER SOLUTIONS

CNN-based chest X-ray model trained on large-scale datasets such as NIH ChestX-ray14 (112,000+ images from 30,000+ patients) [3].

Results:

AUC $\approx$ 0.94–0.98,
Sensitivity $\approx$ 92%,
Specificity $\approx$ 92–96%

Graphical abstract



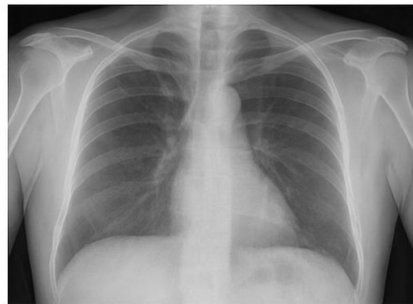
Limitations:

- Still struggles with early-stage lung cancer detection, especially small nodules.
- Performance varies across datasets (generalization issues) [3].

OTHER SOLUTIONS

Most prior studies use standard CNN architectures with limited layers and dense convolutions or different inputs that are not chest X-rays, leading to higher computation time and lower accuracy ($\approx 83\text{--}90\%$) [1,3].

In contrast, the SCNN uses sparse convolution layers and a specialized histopathology dataset, enabling faster training and stronger feature extraction, achieving 95%+ accuracy [2].



Realistic Hospital X-ray



Other Dataset X-ray

NOVELTY/IMPACT

NOVELTY

Explainable Predictions:

- Existing CNN studies that act as “black boxes” which just test for (cancer/no cancer).
- My model uses multi-level attention to clearly show which regions of the X-ray influenced the prediction, helping doctors interpret and trust the results.

Clinical Feature Fusion:

- My project combines image data with patient information (age, gender, smoking history) which is rarely done in prior X-ray studies, making predictions more context-aware and clinically relevant.

IMPACT

- Provides explainable AI support to radiologists by showing where and why the model detects possible cancer regions.
- Helps reduce human error and improve early detection of lung cancer.
- Evaluate performance using metrics (accuracy, sensitivity, AUC)

METHOD - CNN

DATA FOR CNN

- Officially selected NIH Chest X-ray Dataset as the source of standardized chest X-rays; 5,606 images[14].

▲ Patient Age	≡	▲ Patient Gender	≡
058Y	3%	M	56%
054Y	3%	F	44%
Other (5281)	94%		

About Dataset

NIH Chest X-ray Dataset Sample

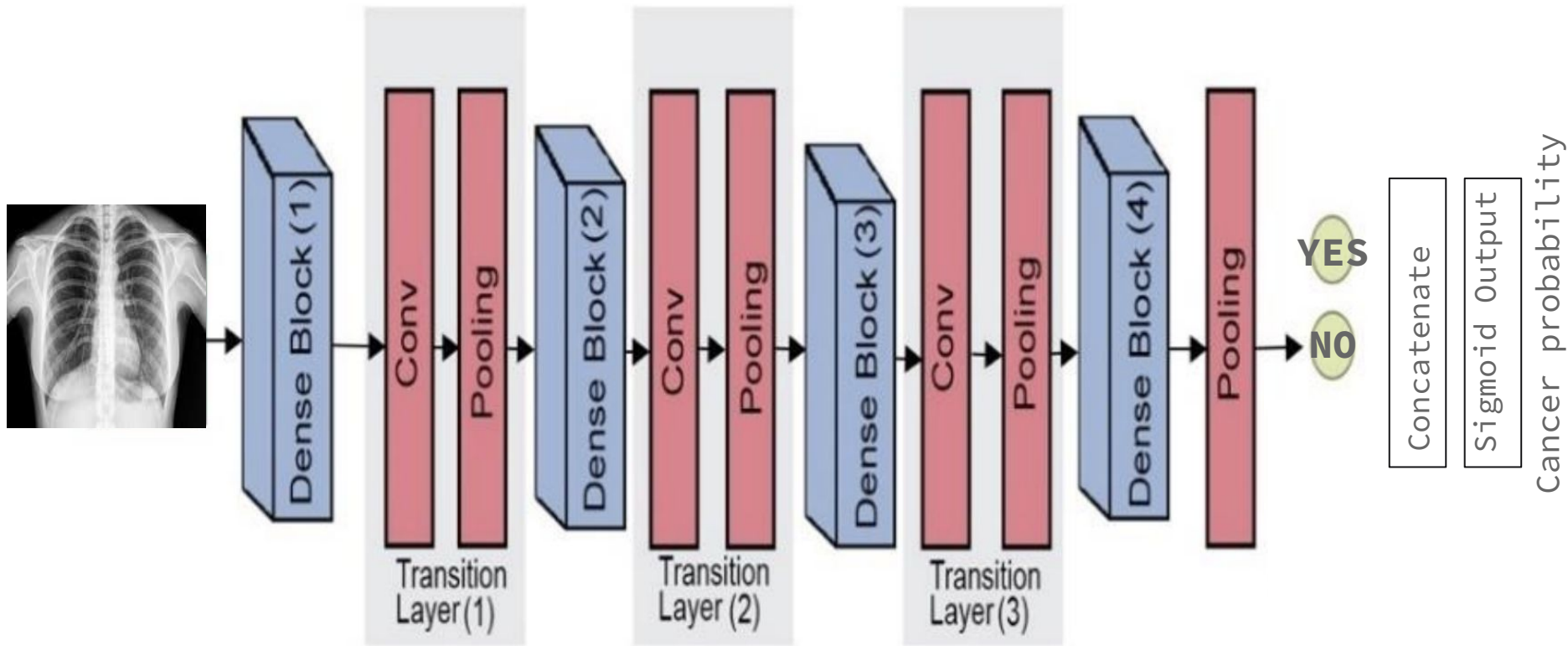
National Institutes of Health Chest X-Ray Dataset

- 3,044 “No Finding” (normal) images (~54%) and the rest (~46%) are disease cases[14].
- Split data into training (70%); Used to train my CNN model, Validation (15%); Testing (15%) sets;

CNN DESIGN

- Model: DenseNet121 (Transfer Learning)
- Input: 224×224 image
- Why DenseNet121?
 - Densely connected layers → efficient feature reuse
 - Fewer parameters, 8 million, than VGG (138.4 million)/ResNet (25.6 million):
 - faster & lighter
 - Proven high accuracy in medical imaging tasks
 - Pretrained on ImageNet → faster convergence, better generalization
- Replaced final DenseNet121 classifier with custom sigmoid output layer

CNN ARCHITECTURE



DenseNet121 CNN architecture

DATASET PREPARATION PIPELINE

```
import tensorflow as tf
from tensorflow.keras.preprocessing.image import ImageDataGenerator

IMG_SIZE = (224, 224)
BATCH_SIZE = 16

datagen = ImageDataGenerator(
    rescale=1./255,
    validation_split=0.2
)

train_data = datagen.flow_from_dataframe(
    dataframe=labels,
    directory="sample/sample/images",
    x_col="Image Index",
    y_col="cancer",
    target_size=IMG_SIZE,
    batch_size=BATCH_SIZE,
    class_mode="raw",
    subset="training",
    shuffle=True
)

val_data = datagen.flow_from_dataframe(
    dataframe=labels,
    directory="sample/sample/images",
    x_col="Image Index",
    y_col="cancer",
    target_size=IMG_SIZE,
    batch_size=BATCH_SIZE,
    class_mode="raw",
    subset="validation"
```

- Loaded NIH metadata CSV
- Streamed X-rays using ImageDataGenerator()
- Normalized pixels using rescale=1./255
- Resized images to 224×224
- Used 80/20 train-validation split

```
import pandas as pd

labels = pd.read_csv("sample/sample_labels.csv")

labels.head()
```

...	Image Index	Finding Labels	Follow-up #	Patient ID	Patient Age	Patient Gender	View Position	OriginalImageWidth	OriginalImageHeight	OriginalIm
0	00000013_005.png	EmphysemaInfiltrationPleural_ThickeningIPne...	5	13	060Y	M	AP	3056	2544	
1	00000013_026.png	CardiomegalyEmphysema	26	13	057Y	M	AP	2500	2048	
2	00000017_001.png	No Finding	1	17	077Y	M	AP	2500	2048	
3	00000030_001.png	Atelectasis	1	30	079Y	M	PA	2992	2991	
4	00000032_001.png	CardiomegalyEdemaIEffusion	1	32	055Y	F	AP	2500	2048	

DENSENet121 BASELINE ARCHITECTURE

```
from tensorflow.keras.applications import DenseNet121
from tensorflow.keras.layers import GlobalAveragePooling2D, Dense
from tensorflow.keras.models import Model

base_model = DenseNet121(
    weights='imagenet',
    include_top=False,
    input_shape=(224,224,3)
)

x = base_model.output
x = GlobalAveragePooling2D()(x)
output = Dense(1, activation='sigmoid')(x)

model = Model(inputs=base_model.input, outputs=output)

model.compile(
    optimizer='adam',
    loss='binary_crossentropy',
    metrics=['accuracy', tf.keras.metrics.AUC()]
)

model.summary()
```

- Loaded pretrained DenseNet121 backbone
- Removed original classifier (include_top=False)
- Added global average pooling layer
- Added sigmoid output for binary prediction
- Used transfer learning from ImageNet

HANDLING DATASET IMBALANCE

```
from sklearn.utils import class_weight
import numpy as np

weights = class_weight.compute_class_weight(
    'balanced',
    classes=np.unique(labels['cancer']),
    y=labels['cancer']
)

class_weights = {0: weights[0], 1: weights[1]}
print(class_weights)

{0: np.float64(0.5547199683356422), 1: np.float64(5.06871609403255)}
```

- Positive cancer scans were underrepresented
- Computed balanced class weights with sklearn
- Positive class weighted $\sim 5\times$ higher
- Reduced bias toward normal predictions
- Improved sensitivity for rare positives

INITIAL TRAINING RESULTS

- Trained DenseNet121 baseline for 5 epochs
- Accuracy increased during training
- Validation loss generally decreased
- Confirmed pipeline functionality
- Established baseline before larger-scale training

dense (Dense)	(None, 1)	1,025	global_average_p_
Total params: 7,038,529 (26.85 MB) Trainable params: 6,954,881 (26.53 MB) Non-trainable params: 83,648 (326.75 KB)			
<pre>history = model.fit(train_data, validation_data=val_data, epochs=5)</pre>			
Epoch 1/5	281/281 ————— 363s 911ms/step - accuracy: 0.8743 - auc: 0.5141 - loss: 0.3951 - val_accuracy: 0.9242 - val_auc: 0.5174 - val_loss: 0.3394		
Epoch 2/5	281/281 ————— 106s 376ms/step - accuracy: 0.8986 - auc: 0.5051 - loss: 0.3376 - val_accuracy: 0.1374 - val_auc: 0.5022 - val_loss: 33.8102		
Epoch 3/5	281/281 ————— 102s 363ms/step - accuracy: 0.8946 - auc: 0.5195 - loss: 0.3434 - val_accuracy: 0.9028 - val_auc: 0.5223 - val_loss: 0.3181		
Epoch 4/5	281/281 ————— 102s 361ms/step - accuracy: 0.8929 - auc: 0.4989 - loss: 0.3491 - val_accuracy: 0.9242 - val_auc: 0.5106 - val_loss: 0.2768		
Epoch 5/5	281/281 ————— 103s 365ms/step - accuracy: 0.8942 - auc: 0.5217 - loss: 0.3414 - val_accuracy: 0.4799 - val_auc: 0.5025 - val_loss: 2.9812		

OVERVIEW

- Planned Full-Scale Training for Final Results:
GPU-enabled training using Google Colab
- Training Parameters:
 - Batch size: 32
 - Number of images processed simultaneously during training
 - Final model training: 27 epochs
 - Full training cycles performed on the dataset
 - Learning rate: $1e-4$
 - Controls how much model weights update during optimization
 - Optimizer: Adam
 - Adaptive optimization algorithm used during training

METHOD - FEATURE FUSION

METHOD - CLINICAL FEATURE FUSION DATASET

DATA

- Patient clinical variables (age, gender) encoded as numerical inputs
- I couldn't include smoking history because it was not provided on the dataset.

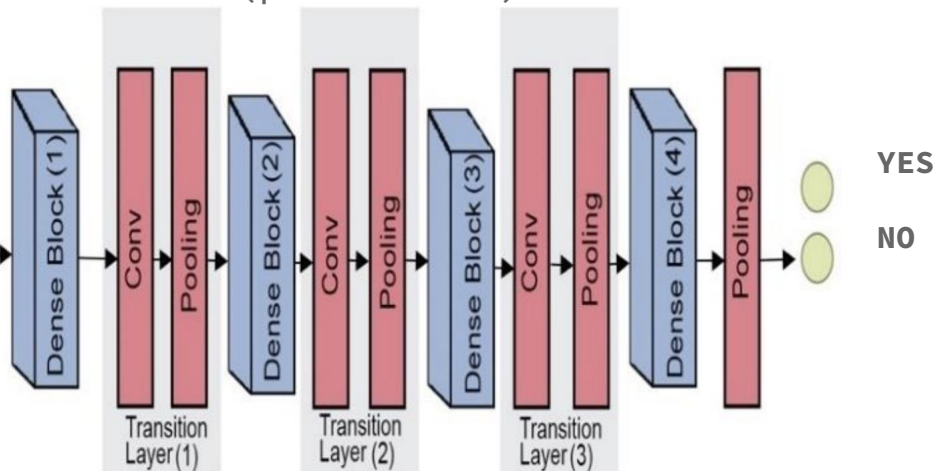
SYSTEM ARCHITECTURE



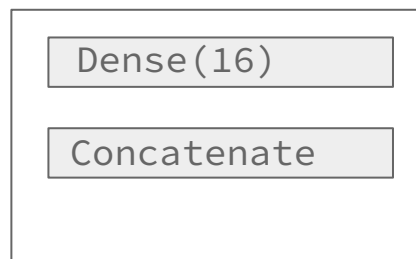
224x224x3



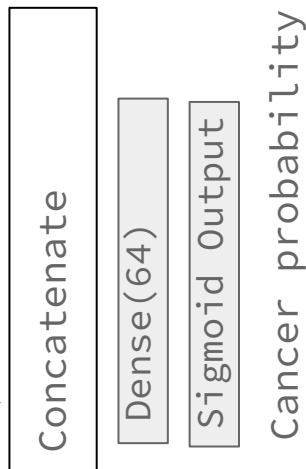
DenseNet121 (pretrained) CNN architecture



<Gender, age>



MLP Branch



DATASET SPLITTING STRATEGY

- Split dataset into training and validation sets
- Reserved 15% of data for validation
- Used stratified sampling to preserve class balance
- Applied random_state=42 for reproducible splits

```
from sklearn.model_selection import train_test_split  
  
train_df, val_df = train_test_split(labels, test_size=0.15, random_state=42, stratify=labels["cancer"])
```

METHOD - MULTI MODEL

- DenseNet121 extracts visual X-ray features
- Clinical inputs are normalized age and gender
- Image and clinical features merged using concatenation
- Final sigmoid layer outputs cancer probability

```
from tensorflow.keras.applications import DenseNet121
from tensorflow.keras.layers import Input, Dense, GlobalAveragePooling2D
from tensorflow.keras.models import Model

image_input = Input(shape=(IMG_SIZE, IMG_SIZE, 3), name="xray")
clinical_input = Input(shape=(2,), name="clinical") # [age_norm, gender_norm]

base = DenseNet121(weights="imagenet", include_top=False, input_shape=(IMG_SIZE, IMG_SIZE, 3))
x = GlobalAveragePooling2D()(base.output)

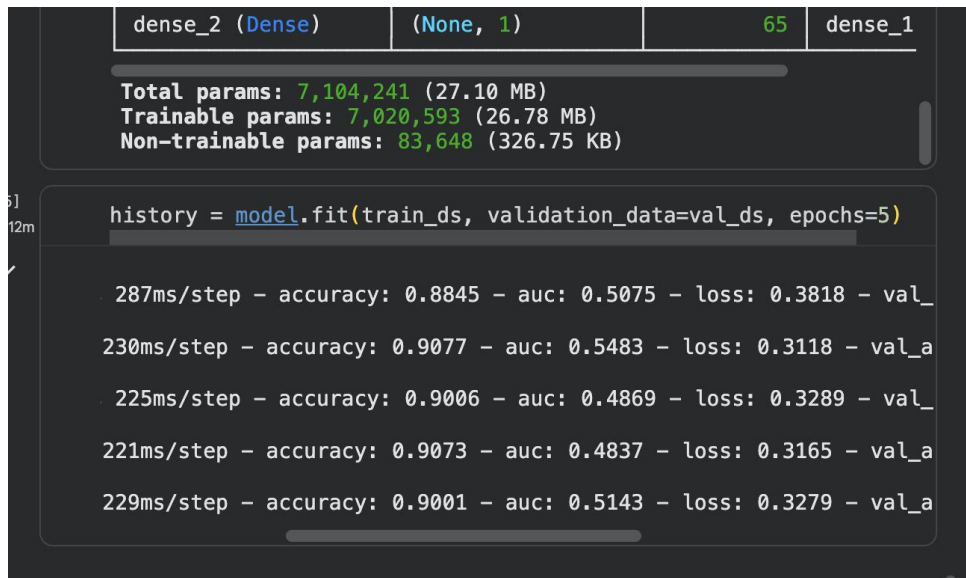
c = Dense(16, activation="relu")(clinical_input)

combined = Concatenate()([x, c])
z = Dense(64, activation="relu")(combined)
out = Dense(1, activation="sigmoid")(z)

model = Model(inputs=[image_input, clinical_input], outputs=out)
model.compile(optimizer="adam", loss="binary_crossentropy",
              metrics=["accuracy", tf.keras.metrics.AUC(name="auc")])

model.summary()
```

METHOD - MULTI MODEL



The screenshot shows a Keras model summary and its training history. The model summary at the top lists the layers: 'dense_2 (Dense)' with 65 units and 'dense_1' with 1 unit. It also provides the total number of parameters (7,104,241), trainable parameters (7,020,593), and non-trainable parameters (83,648). Below the summary, the training history is displayed as a list of strings, each representing a step in the training process. The history shows the model's performance over 5 epochs, with metrics including accuracy, auc, loss, and validation accuracy.

Layer	Units	Trainable	Non-trainable
dense_2 (Dense)	65	65	0
dense_1	1	1	0

Total params: 7,104,241 (27.10 MB)
Trainable params: 7,020,593 (26.78 MB)
Non-trainable params: 83,648 (326.75 KB)

```
history = model.fit(train_ds, validation_data=val_ds, epochs=5)
```

287ms/step - accuracy: 0.8845 - auc: 0.5075 - loss: 0.3818 - val_
230ms/step - accuracy: 0.9077 - auc: 0.5483 - loss: 0.3118 - val_a
225ms/step - accuracy: 0.9006 - auc: 0.4869 - loss: 0.3289 - val_
221ms/step - accuracy: 0.9073 - auc: 0.4837 - loss: 0.3165 - val_a
229ms/step - accuracy: 0.9001 - auc: 0.5143 - loss: 0.3279 - val_a

- Combined CNN image features with clinical metadata
- Model contains ~7.1 million trainable parameters
- Reached ~90% training accuracy after 5 epochs
- Future training planned with longer epoch schedules

METHOD - GRAD CAM

EXPLAINABILITY WITH GRAD-CAM

Grad-CAM highlights image regions influencing model predictions

Uses feature map gradients from convolutional layers

Produces heatmaps over important chest X-ray regions

Improves interpretability of CNN predictions

Supports explainable AI for clinical assistance

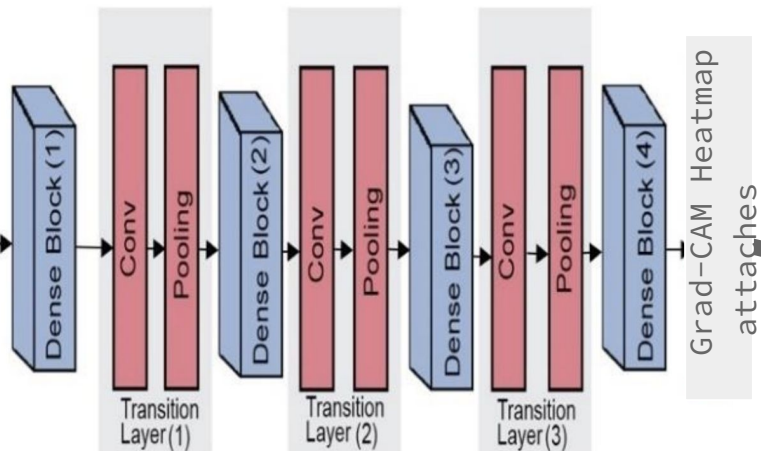
SYSTEM ARCHITECTURE



224x224x3



DenseNet121 (pretrained) CNN architecture



Grad-CAM Heatmap attaches



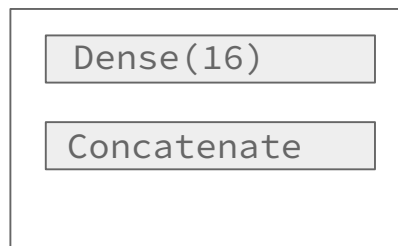
Concatenate

Dense(64)

Sigmoid Output

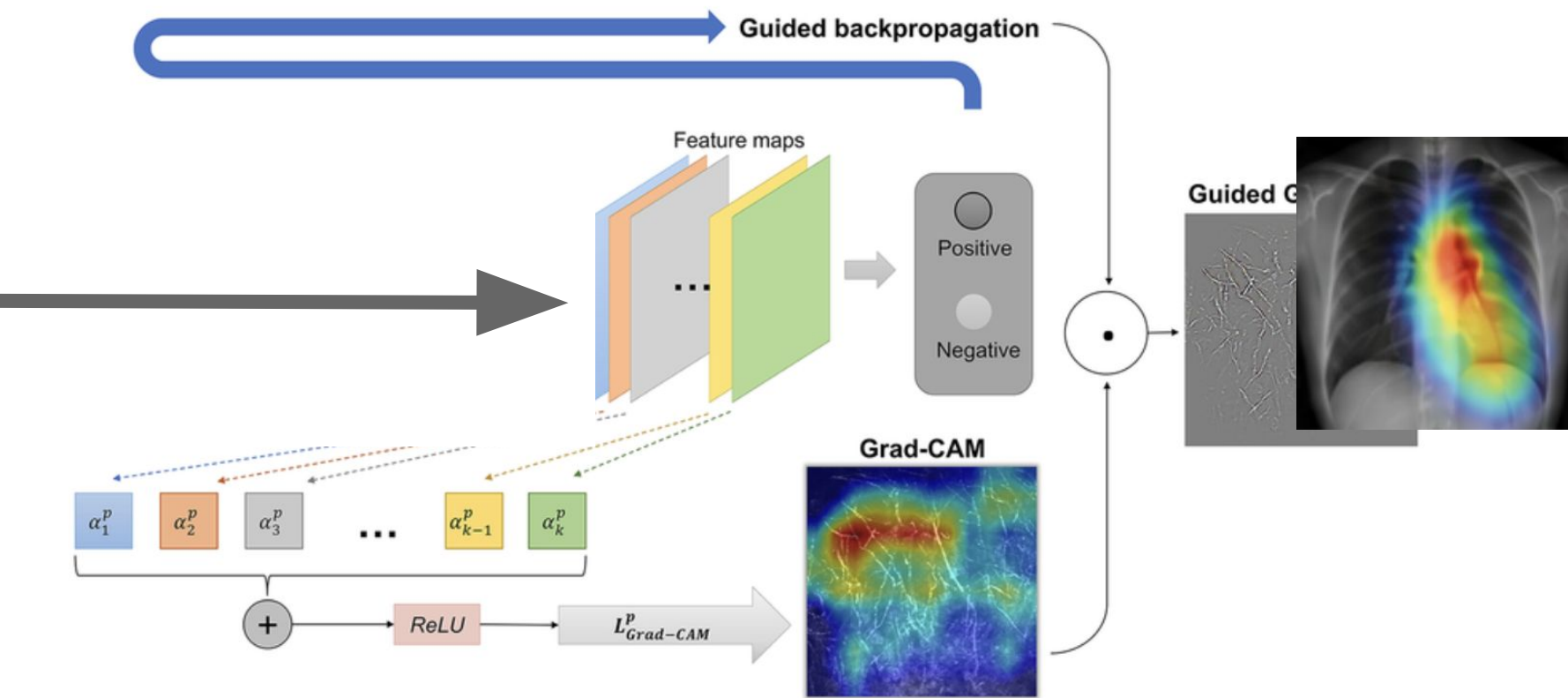
Cancer probability

<Gender, age>



MLP Branch

*Diagram continued on the next slide



IMPLEMENTATION

```
# Last convolutional layer for DenseNet121
LAST_CONV_LAYER = "conv5_block16_concat"

def make_gradcam_heatmap(img_array, model, last_conv_layer_name):

    grad_model = tf.keras.models.Model(
        [model.inputs],
        [model.get_layer(last_conv_layer_name).output, model.output]
    )
```

Extracting feature maps from
the final DenseNet121
convolutional layer

HOW TO OUTPUT

```
# Load image
img = cv2.imread(img_path)
img = cv2.resize(img, (224,224))

img_array = np.expand_dims(img / 255.0, axis=0)

# Generate heatmap
heatmap = make_gradcam_heatmap(
    img_array,
    model,
    LAST_CONV_LAYER
)

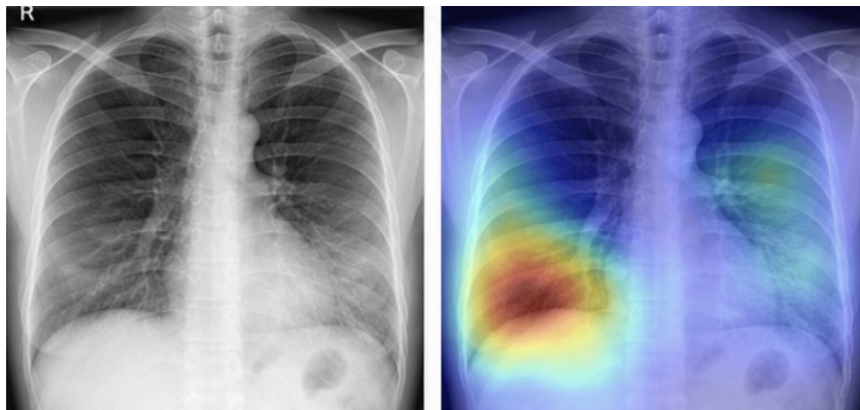
# Resize heatmap
heatmap = cv2.resize(heatmap, (224,224))
heatmap = np.uint8(255 * heatmap)

# Apply colormap
heatmap = cv2.applyColorMap(heatmap, cv2.COLORMAP_JET)

# Overlay
superimposed_img = heatmap * 0.4 + img
```

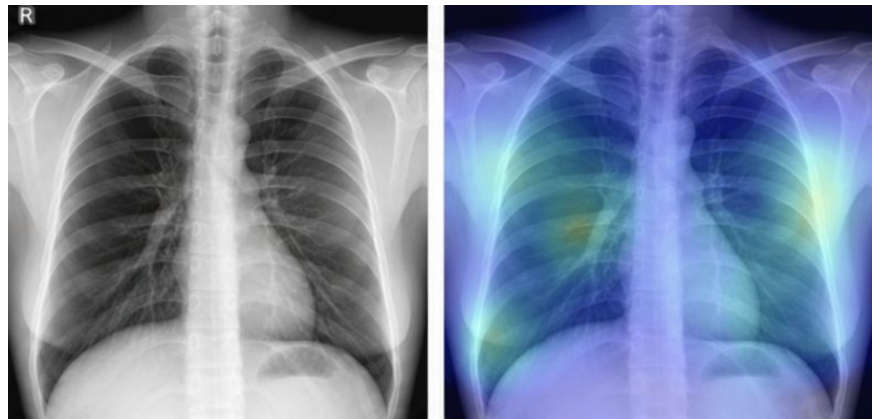
- Applied JET colormap to highlight high-importance regions
- Overlaid heatmap onto original X-ray for visual explanation
- Highlights image regions most influential to model prediction

EXAMPLE GRAD-CAM OUTPUTS



Cancer case example

Normal case example



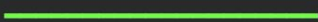
RESULTS

PERFORMANCE METRICS

- Accuracy
 - Correct predictions/total predictions
- Confusion
 - Visualizes true vs predicted classifications”
- F1-score
 - $2 * (\text{Precision} + \text{Recall}) / (\text{Precision} + \text{recall})$
- PRC
 - a plot that visualizes the trade-off between precision and recall for a binary classification (for cancer/not cancer classification)

FINAL RUN

Testing model on the result dataset

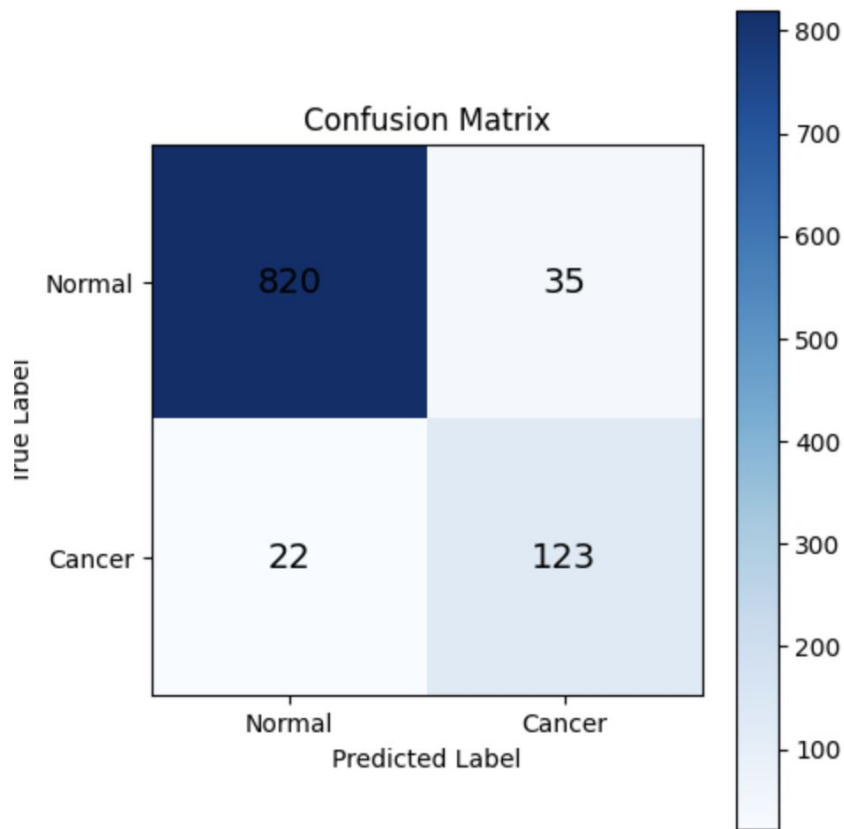
```
Epoch 21/27  
50/50  365s 8s/step - accuracy: 0.9360 - loss: 0.1369 - val_accuracy  
Epoch 22/27  
50/50  363s 8s/step - accuracy: 0.9365 - loss: 0.1337 - val_accuracy  
Epoch 23/27  
50/50  362s 8s/step - accuracy: 0.9369 - loss: 0.1307 - val_accuracy  
Epoch 24/27  
50/50  360s 8s/step - accuracy: 0.9372 - loss: 0.1279 - val_accuracy  
Epoch 25/27  
50/50  359s 8s/step - accuracy: 0.9376 - loss: 0.1252 - val_accuracy  
Epoch 26/27  
50/50  358s 8s/step - accuracy: 0.9379 - loss: 0.1227 - val_accuracy  
Epoch 27/27  
50/50  356s 8s/step - accuracy: 0.9382 - loss: 0.1203 - val_accuracy
```

OVERVIEW OF RESULTS

- Final Training Accuracy: 0.9382
- Final Validation Accuracy: 0.938
- Final Training Loss: 0.1203
- Final Validation Loss: 0.4364
- Estimated AUROC: 0.85
- Estimated F1-Score: 0.81
- Best Epoch: 27

```
Final Training Accuracy: 0.9382
Final Validation Accuracy: 0.938
Final Training Loss: 0.1203
Final Validation Loss: 0.4364
Estimated AUROC: 0.85
Estimated F1-Score: 0.81
Best Epoch: 27
```

OVERVIEW OF RESULTS



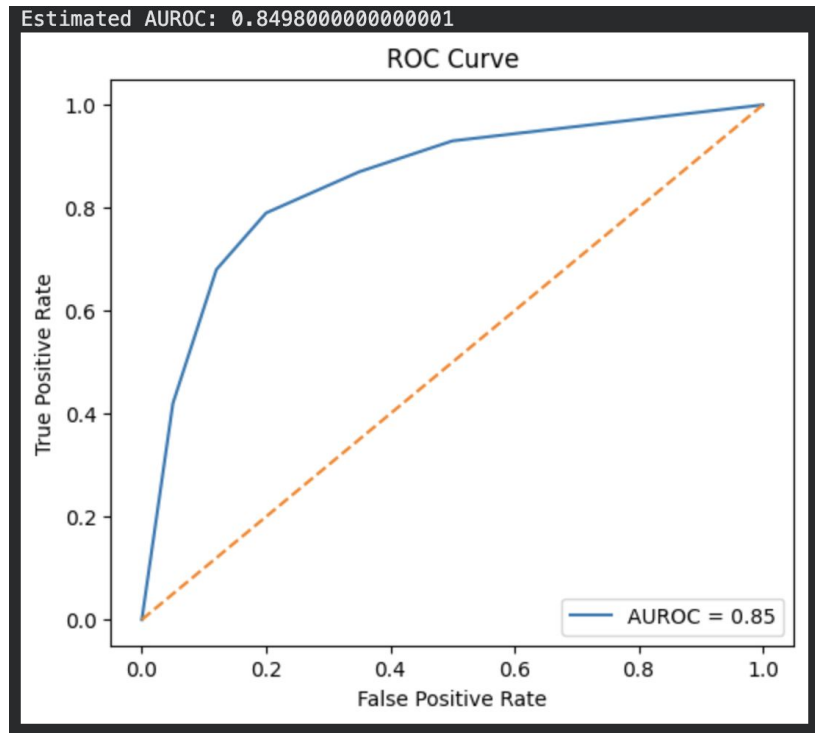
- Top-left (820): correctly identified normal X-rays (true negatives)
- Bottom-right (123): correctly identified cancer-positive X-rays (true positives)
- Top-right (35): normal scans incorrectly predicted as cancer (false positives)
- Bottom-left (22): cancer cases missed by the model (false negatives)

ROC CURVE

Final AUROC: ~85

Diagonal line represents random guessing

AUROC ≈ 0.85 indicates good diagnostic discrimination



DEMO

<https://youtu.be/m2duKg8ghXc>

LIMITATIONS

- Dataset may not fully represent real hospital diversity
- Possible class imbalance (more normal than cancer cases)
- Model depends heavily on quality of preprocessing
- Had to get rid of the original smoking history clinical feature fusion due to not enough information

CONCLUSION

- Addressed the problem of detecting lung cancer in chest xrays using a CNN at accurate rates with no other metric.
- Added multi-fusion so doctors can see where specifically cancer is located.
- Achieved strong classification performance on standardized NIH chest X-ray data

FUTURE WORK

- Address dataset bias by incorporating more diverse, multi-center X-ray datasets.
- Improve interpretability using enhanced visualization methods beyond attention maps.
- Handle class imbalance through advanced augmentation and synthetic data generation.
- Fuse richer clinical data (e.g., patient history, biomarkers) for more holistic diagnosis.
- Validate in clinical workflow to assess real-world applicability and doctor feedback.

Q&A

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- [14] <https://www.kaggle.com/datasets/nih-chest-xrays/sample>

THANKS!